

Livneh Observed vs. Historical GCM - Distributional Shift in BMU Frequencies
Earth Mover's Distance permutation test (n = 5000), per-model

	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
ACCESS-CM2	0.757	0.444	0.736	0.626	0.809	0.192	0.579	0.385
CESM2-LENS	0.080	0.411	0.018*	0.230	0.636	0.153	0.643	0.603
CNRM-ESM2-1	0.684	0.437	0.394	0.540	0.545	0.384	0.595	0.550
EC-Earth3	0.331	0.531	0.343	0.258	0.931	0.640	0.124	0.391
EC-Earth3-Veg	0.525	0.607	0.546	0.170	0.872	0.405	0.530	0.266
FGOALS-g3	0.452	0.313	0.738	0.265	0.859	0.790	0.789	0.157
GFDL-ESM4	0.492	0.531	0.709	0.834	0.831	0.615	0.770	0.797
HadGEM3-GC31-LL	0.797	0.998	0.413	0.960	0.551	0.484	0.438	0.156
INM-CM5-0	0.905	0.440	0.890	0.990	0.841	0.304	0.762	0.295
IPSL-CM6A-LR	0.492	0.891	0.134	0.640	0.982	0.570	0.447	0.478
KACE-1-0-G	0.522	0.090	0.709	0.567	0.810	0.006**	0.743	0.825
MIROC6	0.685	0.798	0.659	0.812	0.210	0.896	0.711	0.497
MPI-ESM1-2-HR	0.214	0.759	0.349	0.401	0.620	0.285	0.908	0.606
MRI-ESM2-0	0.097	0.522	0.930	0.961	0.700	0.777	0.448	0.328
TaiESM1	0.614	0.770	0.387	0.869	0.477	0.763	0.837	0.584

